Comparative Research on GPU Operating Systems and Their Application in AWS Cloud Platforms

Student Name:

Student ID: U2865199

Module Code: CN7029

Module Title: Computer Architecture and Networks

School: School of Architecture, Computing & Engineering

Table of Contents

1. Introduction....3

2. Task 1: GPU Operating Systems and Command Demonstration....4

2.1 Overview of GPU Operating Systems....4

2.2 Command 1: GPU Detection / Enumeration....5

2.3 Command 2: GPU Driver and Kernel Module Information....6

2.4 Command 3: GPU Memory and Resolution....6

2.5 Command 4: GPU Driver Version and OS Architecture Mapping....7

2.6 Command 5: GPU Resource Utilisation....8

2.7 Summary of Task 1....9

3. Task 2: Cloud Computing Platform Implementation Using AWS EC2....9

3.1 Overview of AWS EC2....9

3.2 EC2 Main Page and Instance Configuration....9

3.3 Instance Type Selection....10

3.4 Key Pair Creation and Network Settings....11

3.5 Storage Configuration and Instance Launch....12

3.6 Monitoring, Stopping, and Deleting the Instance....13

3.7 Summary of Task 2....15

4. Task 3: Reflective Evaluation....15

4.1 Role of Operating Systems in GPU Computing....16

4.2 GPU Operating Systems in Cloud Environments....16

4.3 Comparison of Local and Cloud GPU Management....16

4.4 Critical Evaluation of AWS Implementation....16

4.5 Learning Outcomes....17

5. Task 4: Conclusion....17

5.1 Summary of Work Completed....17

5.2 Key Findings....17

5.3 Limitations....18

5.4 Future Work....18

Reference....19

List of Figures:

Figure 1 GPU Detection / Enumeration....6

Figure 2 GPU Driver and Info....6

Figure 3 GPU Memory and Resolution....7

Figure 4 GPU Driver Version & OS Architecture Mapping....8

Figure 5 GPU Resource Utilisation....8

Figure 6 Ec2 main page....10

Figure 7 instance config....10

Figure 8 Instance type....11

Figure 9 Key pair creation....12

Figure 10 Network setting....12

Figure 11 Storage config....13

Figure 12 Launch instance status....13

Figure 13 Monitoring section....14

Figure 14 Stopping ec2 instance....14

Figure 15 Instance deletion....15

1. Introduction

Graphics Processing Units (GPUs) are essential elements of contemporary computing architectures because they are capable of computing large-scale parallel programs fast. Previously owned by the computation of graphics quickly, GPUs are being adopted with high-performance computing, artificial intelligence, machine learning, data analysis, and cloud services. They are much more efficient on data-intensive workloads, as their architecture has a large number of cores able to execute a large number of threads concurrently, unlike their traditional Central Processing Unit counterparts (CPUs).

The successful operation of the GPUs is influenced by its operating system (OS), which performs the role of hardware abstraction, driver administration, memory allocation, scheduling, and secure access to the GPUs. The modern operating systems (like Windows and Linux) include the mechanisms for managing the GPU, which enable applications to gain access to the hardware of the GPU without necessarily engaging with the lower-level components (Yu *et al.*, 2022). Besides that, the Amazon Web Services (AWS) and Microsoft Azure cloud computing infrastructure offer a means to VM-scale the access to the powerful computing capabilities and conduct the on-demand access to the high-powered computing resources.

This assignment will be a comparative and practical study of the working of the GPUs' operating system and how they can be used in the cloud platform, particularly in AWS. The report is divided into four tasks. Task 1 provides an example to show the use of the commands in the operating system of the GPU to find, configure, and monitor GPU resources. Task 2 records the practicable adoption of a cloud computing platform situated under AWS EC2. Task 3 offers a reflective assessment of the support of operating systems on the use of graphic cards in the local environment and the cloud environment. Task 4 is the final part of the study, as it summarizes some of the main findings and insights that were gotten during the coursework.

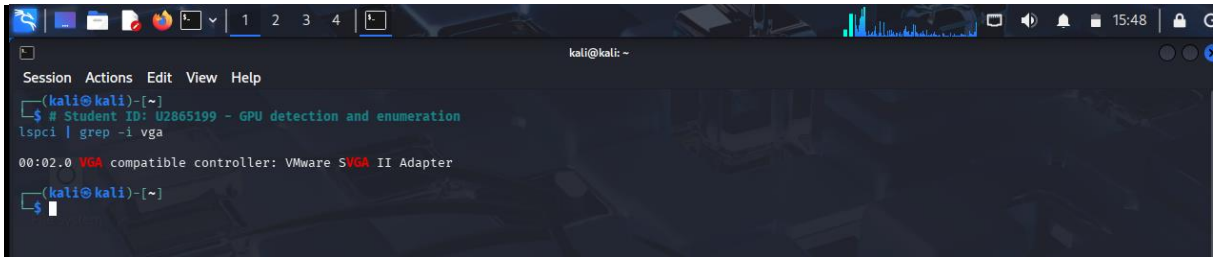
2. Task 1: GPU Operating Systems and Command Demonstration

2.1 Overview of GPU Operating Systems

[redacted]

[redacted]

[redacted]



[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
┌──(kali㉿kali)-[~]
└─$ # Student ID: U2865199 - GPU memory and resolution
xdpyinfo | grep -i dimensions

  dimensions:    1280×800 pixels (338×211 millimeters)

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ # Student ID: U2865199 - GPU driver version and OS architecture mapping
modinfo vmwgfx | grep -E 'version|description|author'
uname -a

version:        2.21.0.0
description:    Standalone drm driver for the VMware SVGA device
author:         VMware Inc. and others
srcversion:     E66A71A504D1EED89C66F0B
vermagic:       6.16.8+kali-amd64 SMP preempt mod_unload modversions
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64 GNU/Linux

┌──(kali㉿kali)-[~]
└─$
```

[redacted]

[redacted]

[redacted]

```
┌──(kali㉿kali)-[~]
└─$ # Student ID: U2865199 - GPU/graphics utilisation monitoring via kernel
top -b -n 1 | head -n 15

top - 15:57:27 up 14 min,  1 user,  load average: 0.21, 0.31, 0.18
Tasks: 166 total,   1 running, 165 sleeping,   0 stopped,   0 zombie
%Cpu(s):  8.0 us,  8.0 sy,  0.0 ni, 80.0 id,  4.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1971.1 total,    772.3 free,    835.0 used,    513.9 buff/cache
MiB Swap:   1906.0 total,   1906.0 free,      0.0 used.   1136.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6205 kali      20   0   10416   5252   3200 R  27.3   0.3   0:00.03 top
      1 root      20   0   24476  15168  10840 S   0.0   0.8   0:01.95 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_wo+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+

┌──(kali㉿kali)-[~]
└─$
```

[redacted]

[redacted]

[redacted]

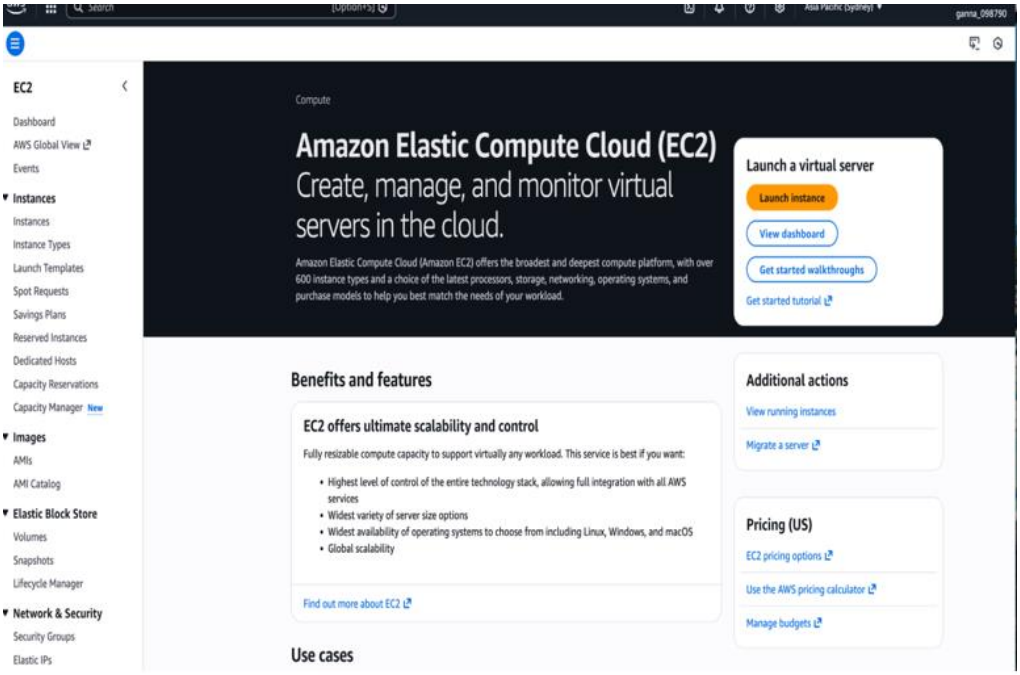
[redacted]

[redacted]

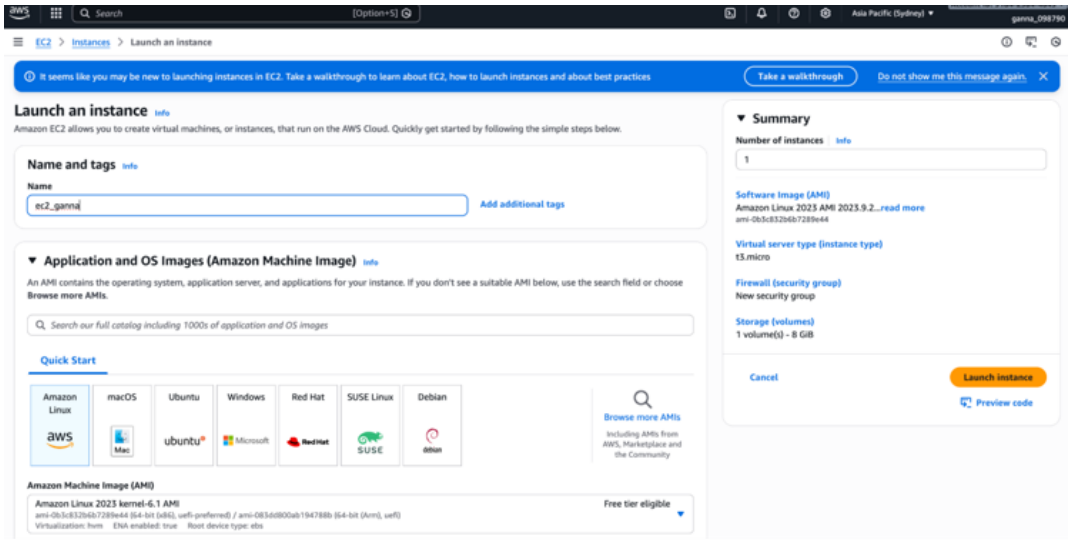
[redacted]

[redacted]

[redacted]



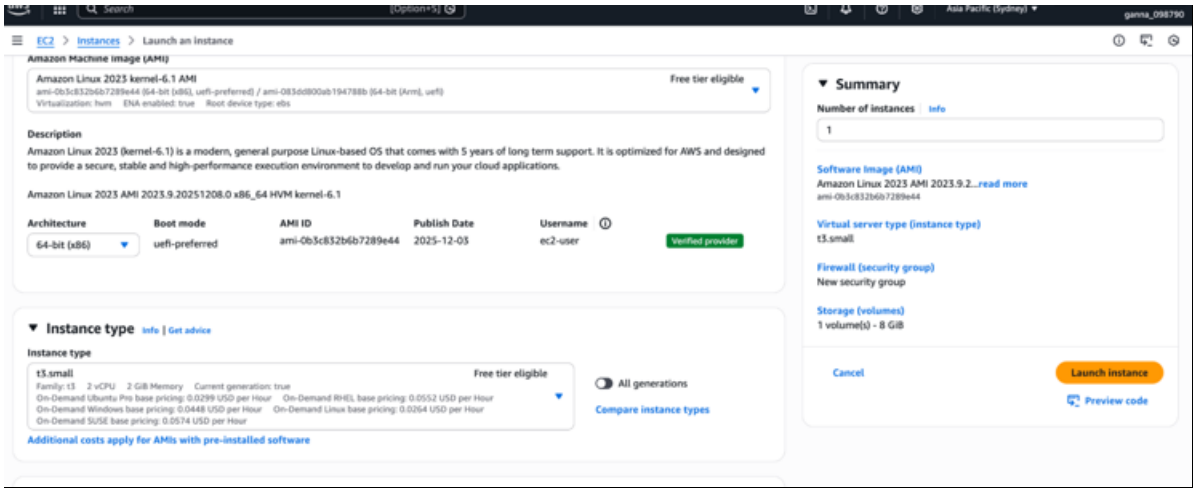
[redacted]



[redacted]

[redacted]

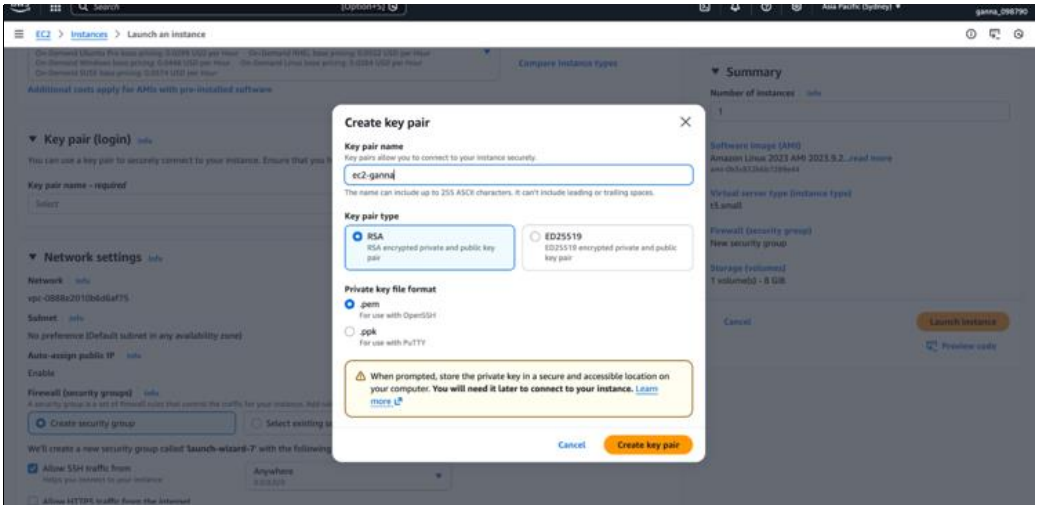
[redacted]



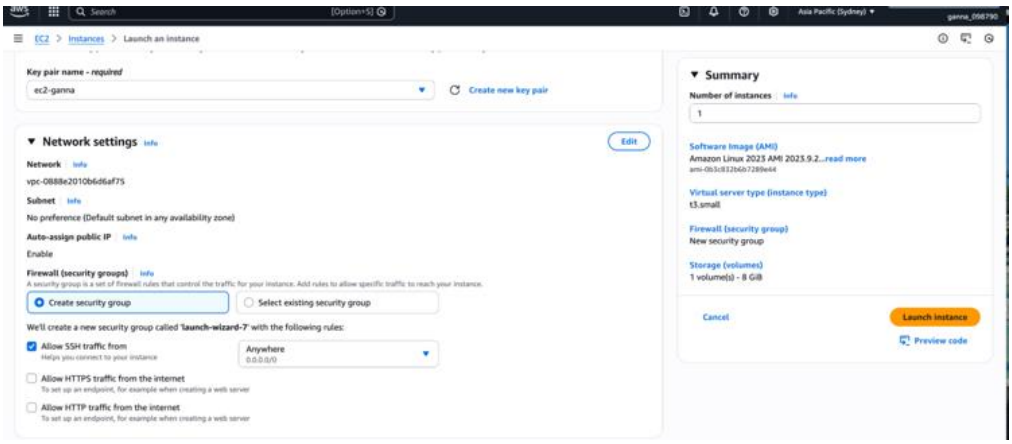
[redacted]

[redacted]

[redacted]



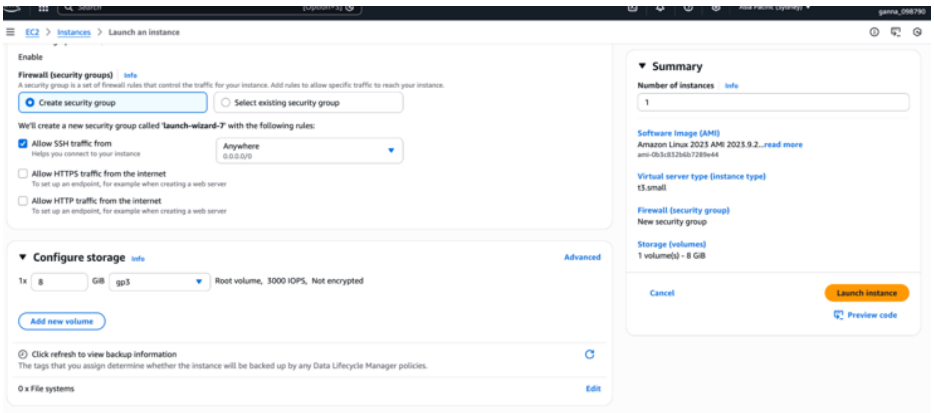
[redacted]



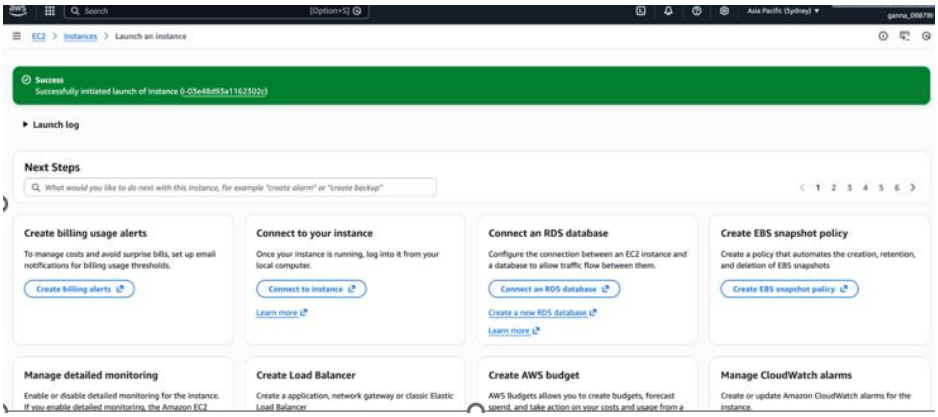
[redacted]

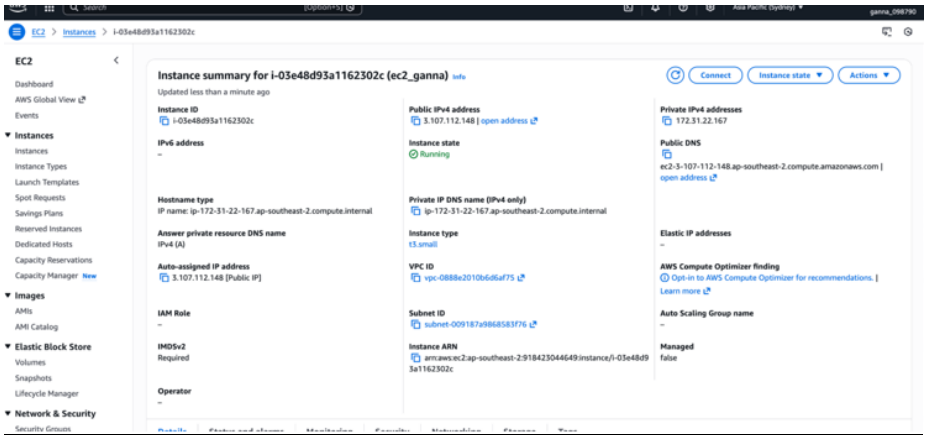
[redacted]

[redacted]



[redacted]





[redacted]

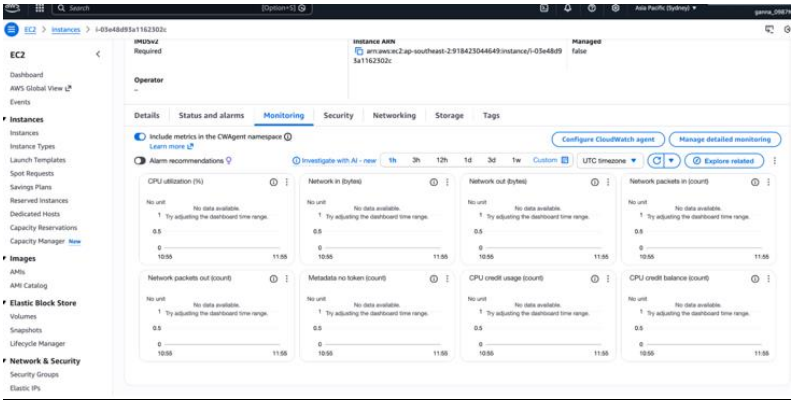
[redacted]

[redacted]

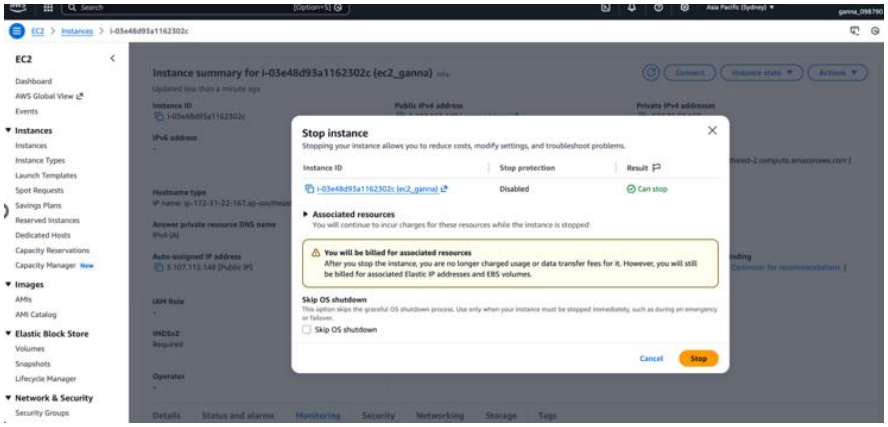
[redacted]

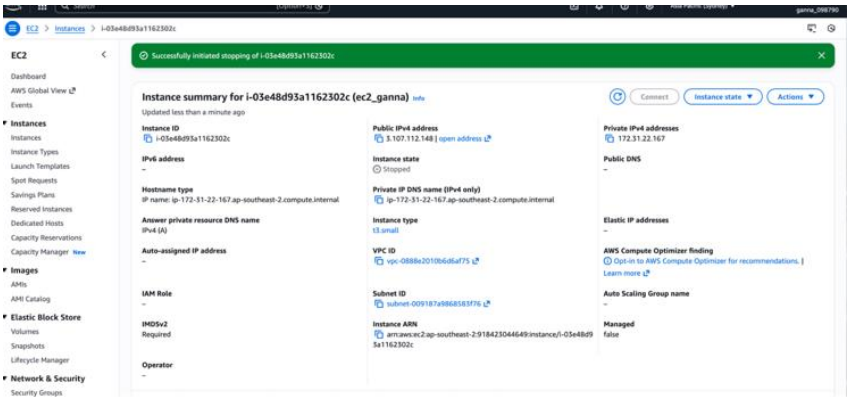
[redacted]

[redacted]

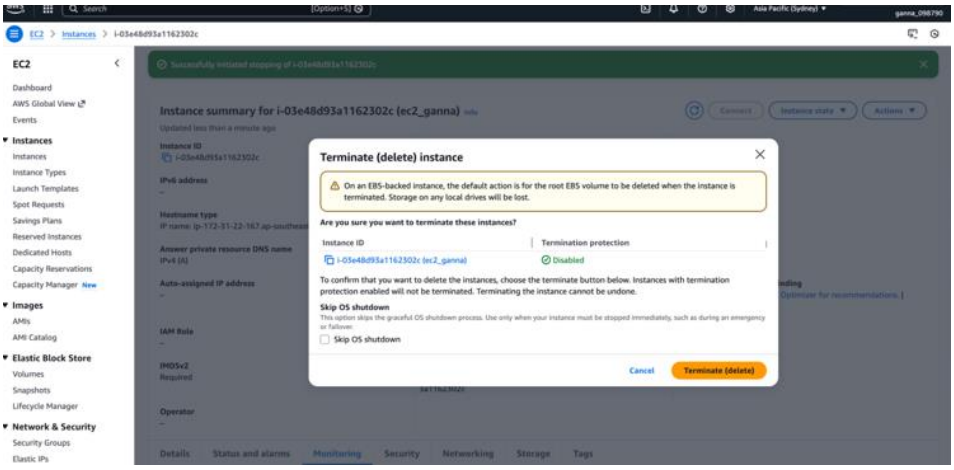


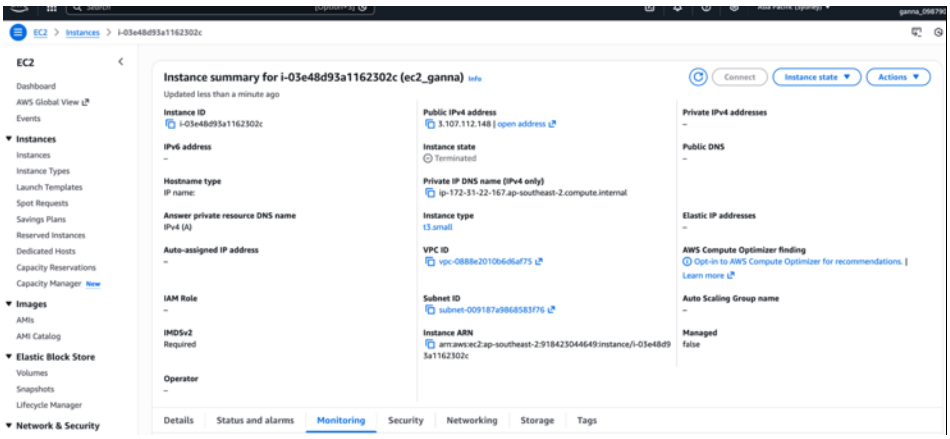
[redacted]





[redacted]





[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]